

41052 - Advanced Algorithms  
Programming Assignment 1 - Fast Fourier Transform

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**Contents**

|          |                                              |          |
|----------|----------------------------------------------|----------|
| <b>1</b> | <b>A little history</b>                      | <b>2</b> |
| <b>2</b> | <b>Relevant background knowledge</b>         | <b>2</b> |
| <b>3</b> | <b>Fast Fourier Transform Implementation</b> | <b>2</b> |
| <b>4</b> | <b>The Empirical Study</b>                   | <b>2</b> |
| <b>5</b> | <b>Lesson learned</b>                        | <b>2</b> |
| <b>6</b> | <b>AI usage Declaration</b>                  | <b>2</b> |

## 1 A little history

In 1802, Napoleon Bonaparte appointed Joseph Fourier as the Prefect (top regional administrator) of the Department of Isère in Grenoble, France. There, Fourier was tasked with managing road construction and draining swamps by day. At night, Fourier spent his time investigating thermodynamics. He started by working from Newton's Law of Cooling, and eventually developed a method to convert discrete harmonic sums into a continuous integral transform pair. He presented his work in 1807 to the Institut de France, but his work was stalled by the review committee, which included Lagrange and Laplace. It is not until 1822, when Fourier was elected as Permanent Secretary of the Académie des Sciences, he bypassed the peer review process and formally introduced what would be known as the *Fourier Transform* today.

The Fourier Transform was used in the continuous domain, which is not relevant to this project. The discrete version actually predate this and was not invented by Fourier! In 1754, while computing for the orbit of celestial bodies, Alexis Clairaut published what is widely considered the earliest discrete Fourier-like formula. In 1805, Carl Friedrich Gauss polished this work and created the complete *Discrete Fourier Transform* and the *radix-2 Fast Fourier Transform (FFT)* to compute DFT faster. All of this was to help him to interpolate the orbits of asteroids.

In the 20<sup>th</sup> century, as electrical engineers and numerical analysts adapted Fourier analysis for sampled signals and digital computers, they applied Fourier's conceptual frequency-domain framework to discrete arrays. The mathematical operation was Gauss's discrete trigonometric interpolation, but it was renamed the Discrete Fourier Transform to reflect its role as the discrete counterpart to Fourier's continuous theory.

## 2 Relevant background knowledge

## 3 Fast Fourier Transform Implementation

## 4 The Empirical Study

## 5 Lesson learned

## 6 AI usage Declaration